

Insup Lee

Cybersecurity & AI Researcher

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Research Interests

My primary research focuses on **AI for cybersecurity and threat intelligence**. I have developed **domain-specific generative models** that preserve structural properties downstream classifiers require, ranging from continuous physical signals in wireless communications to discrete TTP sequences in cyber threat analysis. I recently expanded this foundation to LLMs for cyber threat knowledge graph extraction, shifting my focus from synthetic data generation to automated intelligence extraction. To advance this trajectory, I intend to deploy **agentic AI systems** to automate complex security tasks and investigate their vulnerabilities, contributing to the field of **securing AI**.

Education

Ph.D. Candidate in Cybersecurity, Korea University – Seoul, Republic of Korea Sep 2019 – Present

- Advisors: Prof. Sangjin Lee and Prof. Seokhie Hong | Expected Graduation: Aug 2026
- Dissertation: Domain-Specific Generative Models for Data Augmentation in Cybersecurity

B.E. in Cyber Defense, Korea University – Seoul, Republic of Korea Mar 2014 – Feb 2018

Work Experience

Lecturer, Korea University – Seoul, Republic of Korea Sep 2025 – Feb 2026

- Taught graduate-level course “Computer Networks”

Research Engineer (Cyber Officer), Ministry of National Defense – Republic of Korea Aug 2023 – May 2025

- Led international joint research on AI-based security with the UAE Ministry of Defense as PI ([J6])
- Taught cybersecurity course “Penetration Testing” to UAE cyber defense personnel

Researcher, Agency for Defense Development (ADD) – Seoul, Republic of Korea Jul 2018 – Jul 2023

- Analyzed nation-state APT campaigns using MITRE ATT&CK, developing NLP-based methods for TTP sequence analysis and attack group attribution ([J1]–[J3], [J5])
- Applied generative models (GANs, sequence models) for data augmentation in anomaly detection ([C1], [C2])

Selected Publications

- [TIFS’25] **I. Lee et al.**, “MuCamp: Generating Cyber Campaign Variants via TTP Synonym Replacement for Group Attribution”
- [TDSC’24] **I. Lee et al.**, “UniQGAN: Towards Improved Modulation Classification With Adversarial Robustness Using Scalable Generator Design”
- [SPL’25] **I. Lee et al.**, “Enhancing Modulation Classification via Diffusion Transformers for Drone Video Signal Processing”
- [CAL’26] **I. Lee et al.**, “LeakDiT: Diffusion Transformers for Trace-Augmented Side-Channel Analysis”

Full Publications

International Publications (C: Conference, J: Journal)

C4 *Mnemonic Hack: Recovering the Master Seed from Bitcoin Hardware Wallets via Side-Channel Attacks*
J. Baek, G. Ahn, S. Park, D. Bae, G. Kim, **I. Lee**, S. Hong, and H. Kim
ACM Conference on Computer and Communications Security (CCS’26), Nov. 2026.

- C3 *Exploiting Per-Core Leakage: Electromagnetic Side-Channel Monitoring of Multicore Architectures*
D. Bae, S. Park, **I. Lee**, Y. Jung, K. Lee, H. Kim, and S. Hong
ACM/IEEE Design Automation Conference (DAC'26), Jul. 2026.
- J8 *LeakDiT: Diffusion Transformers for Trace-Augmented Side-Channel Analysis*
I. Lee, D. Bae, S. Hong, and S. Lee
IEEE Computer Architecture Letters, Vol. 25, No. 1, pp. 5-8, Jan./Jun. 2026.
- J7 *Multi-Step LLM Pipeline for Enhancing TTP Extraction in Cyber Threat Intelligence*
H. Kim, D. Lee, **I. Lee**, S. Lee, and S. Lee
IEEE Access, Vol. 13, pp. 179696-179710, Oct. 2025.
- J6 *Enhancing Modulation Classification via Diffusion Transformers for Drone Video Signal Processing*
I. Lee, K. Alteneiji, and M. Alghfeli
IEEE Signal Processing Letters, Vol. 32, pp. 3325-3329, Aug. 2025.
- J5 *MuCamp: Generating Cyber Campaign Variants via TTP Synonym Replacement for Group Attribution*
I. Lee and C. Choi
IEEE Transactions on Information and Forensics Security (TIFS), Vol. 20, pp. 6162-6174, Jun. 2025.
- J4 *UniQGAN: Towards Improved Modulation Classification With Adversarial Robustness Using Scalable Generator Design*
I. Lee and W. Lee
IEEE Transactions on Dependable and Secure Computing (TDSC), Vol. 21, No. 2, pp. 732-745, Mar./Apr. 2024.
- J3 *Camp2Vec: Embedding Cyber Campaign With ATT&CK Framework for Attack Group Analysis*
I. Lee and C. Choi
ICT Express, Vol. 9, No. 6, pp. 1065-1070, Dec. 2023.
- J2 *Exploiting TTP Co-occurrence via GloVe-Based Embedding With ATT&CK Framework*
C. Shin, **I. Lee**, and C. Choi
IEEE Access, Vol. 11, pp. 100823-100831, Sep. 2023.
- J1 *BAN: Predicting APT Attack Based on Bayesian Network With MITRE ATT&CK Framework*
Y. Kim, **I. Lee**, H. Kwon, G. Lee, and J. Yoon
IEEE Access, Vol. 11, pp. 91949-94968, Aug. 2023.
- C2 *Anomaly Dataset Augmentation Using Sequence Generative Models*
S. Shin, **I. Lee**, and C. Choi
IEEE International Conference on Machine Learning and Applications (ICMLA'19), Dec. 2019.
- C1 *Opcode Sequence Amplifier Using Sequence Generative Adversarial Networks*
C. Choi, S. Shin, and **I. Lee**
International Conference on ICT Convergence (ICTC'19), Oct. 2019.

Domestic Publications (Korean)

(†: Corresponding Author)

- S. Park, D. Bae, **I. Lee**, H. Kim, and S. Hong, "EM-Based Anomaly Detection using a Dual-Domain Approach," in Proc. of the KIISC Winter Conference (CISC-W), Nov. 2025. (Selected as an Outstanding Paper Award)
- H. Park and **I. Lee**[†], "Enhanced DDoS Detection via Traffic Volume-Based Labeling and Transfer Learning," Journal of Internet Computing and Services (JICS), Vol. 26, No. 4, pp. 1-8, Aug. 2025.
- K. Kim and **I. Lee**[†], "User Behavior Embedding via TF-IDF-BVC for Web Shell Detection," Journal of The Korea Institute of Information Security & Cryptology (JKIISC), Vol. 34, No. 6, pp. 1231-1238, Dec. 2024.

Patents

- C. Choi and **I. Lee**, "Method for Augmentating Cyber Attack Campaign Data to Identify Attack Group, and Security," Korea Patent Application Number. 10-2024-0176082, December 2, 2024.
- C. Choi, **I. Lee**, C. Shin, and S. Lee, "Information Identification Method and Electronic Apparatus Thereof," Korea Patent Application Number. 10-2024-0006106, January 15, 2024.
- C. Choi, C. Shin, S. Shin, S. Seo, and **I. Lee**, "Method for Training Attack Prediction Model and Device Therefor," U.S. Patent Application Number. 18/126,005; U.S. Patent Number. US20230308462A1, September 28, 2023.
- C. Choi, S. Shin, and **I. Lee**, "Apparatus, Method, Computer-readable Storage Medium and Computer Program for Generating Operation Code," Korea Patent Application Number. 10-2019-0141865, November 07, 2019; Korea

Patent Number. 10-2246797, April 30, 2021.

Other Experience

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| AI Cyber Challenge (AIXCC), DARPA and ARPA-H, USA | Apr 2024 – Aug 2024 |
| <ul style="list-style-type: none">• Participated in the semifinal round as a member of Team KORIA, submitting our cyber reasoning system that leverages LLMs for automated detection and patching of software vulnerabilities | |
| SW Outsourcing Development, KCMVP-Certified Cryptographic Module | Jun 2017 – May 2018 |
| <ul style="list-style-type: none">• Implemented a cryptographic module with 25,000 LoC in C - ARIA block cipher (modes: ECB, CBC, CTR), hash functions (SHA-256, SHA-512), and HMAC-based DRBG for Windows (.dll) and Linux (.so) | |

Awards and Honors

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| • KU Graduate School Achievement Award
Korea University, Seoul, Republic of Korea | Feb 2026 |
| • Outstanding Paper Award
CISC-W'25, KIISC | Nov 2025 |
| • Certificate of Commendation (UAE-ROK Engagement Program)
United Arab Emirates Ministry of Defense | Mar 2025 |
| • Ambassador's Commendation
Embassy of the Republic of Korea to the United Arab Emirates | Mar 2025 |
| • Full Tuition Scholarship (Korea University)
Ministry of National Defense, Republic of Korea | Mar 2014 – Feb 2018 |

Professional Service

Reviewer

- IEEE Transactions on Dependable and Secure Computing (TDSC), 2025
- IEEE Transactions on Information Forensics and Security (TIFS), 2026
- IEEE Transaction on Communications (TCOM), 2025, 2026
- IEEE Journal on Selected Areas in Communications (JSAC), 2025, 2026

Mentoring Experience

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| • Sujin Park (Ph.D. Student at Korea University)
Mentored research on side-channel analysis for anomaly detection, resulting in two co-authored domestic conference publications | Jun 2025 – Present |
| • Hyoungrok Kim (M.S. Student at Korea University)
Advised research on LLM-based TTP extraction for cyber threat intelligence, resulting in a journal publication at IEEE Access | Mar 2025 – Sep 2025 |
| • Hyunjun Park (Navy Lieutenant at Ministry of National Defense)
Supervised research on DDoS detection via transfer learning, resulting in a domestic journal publication as corresponding author | Nov 2024 – Feb 2025 |
| • Kangmun Kim (First Lieutenant at Cyber Operations Command)
Supervised research on web shell detection via user behavior embedding, resulting in a domestic journal publication as corresponding author | Jan 2024 – Sep 2024 |

Reference

Sangjin Lee

Professor, School of Cybersecurity, Korea University

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Seokhie Hong

Former Professor, School of Cybersecurity, Korea University

Relationship: Ph.D. Advisor

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Relationship: Research Mentor (Former PI at ADD)

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Jongin Lim

Professor Emeritus, School of Cybersecurity, Korea University

Former Special Presidential Advisor for Cybersecurity, Republic of Korea

Relationship: Undergraduate Advisor

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